WEEK 2: EXPLORATORY DATA ANALYSIS (EDA)

1. Introduction

Exploratory Data Analysis (EDA) is an important step in the data science process. It involves examining, summarizing, and visualizing a dataset to understand its structure, identify patterns, detect potential outliers, and discover relationships between variables.

During Week 2 of the Data Science Internship, Exploratory Data Analysis was performed on the Wine Recognition dataset. The analysis focused on understanding the distribution of wine classes, examining numerical features, checking data quality, exploring correlations, and conducting a statistical hypothesis test.

Python libraries, including Pandas, NumPy, Matplotlib, Seaborn, and SciPy, were used to perform the analysis and visualize the results.

2. Objectives

The main objectives of this task were:

- To understand the structure and dimensions of the Wine Recognition dataset.
- To examine missing values and duplicate records.
- To calculate descriptive statistics for numerical features.
- To analyze the distribution of wine classes.
- To visualize feature distributions using histograms.
- To identify potential outliers using the Interquartile Range (IQR) method.
- To investigate relationships between numerical variables using scatter plots and correlation analysis.
- To compare alcohol content across different wine classes.
- To conduct a one-way ANOVA test to determine whether the mean alcohol content differs significantly among wine classes.

3. Dataset Description

The Wine Recognition dataset was used for this analysis. It contains numerical measurements of wine samples belonging to three different wine classes.

The dataset is commonly used for classification and statistical analysis because its numerical features can help distinguish between wine classes.

Dataset Summary

Property	Description
Dataset Name	Wine Recognition Dataset
Total Records	178
Total Columns	14
Numerical Predictor Features	13
Target Variable	wine_class
Number of Classes	3
Missing Values	0
Duplicate Records	0

The dataset contains 13 predictor features and one target column, wine_class, which identifies the class of each wine sample.

Wine Class Distribution

The dataset contains the following number of samples:

Wine Class	Number of Samples
Class 0	59
Class 1	71
Class 2	48
Total	**178**

The class distribution shows that Class 1 has the highest number of samples, while Class 2 has the lowest. However, all three classes are represented in the dataset.

4. Tools and Technologies Used

The following tools and Python libraries were used:

- **Python:** Programming language used to perform data analysis.
- **Jupyter Notebook:** Environment used to execute code and document the analysis.

- **Pandas:** Used for data loading, manipulation, grouping, and statistical summaries.
- **NumPy:** Used for numerical operations.
- **Matplotlib:** Used to create charts and visualizations.
- **Seaborn:** Used to generate the correlation heatmap.
- **SciPy:** Used to perform the one-way ANOVA statistical test.

5. Methodology

The analysis was performed through the following steps.

5.1 Data Loading

The cleaned dataset was loaded into a Pandas DataFrame using the read_csv() function.

The first few rows were examined using head() to verify that the dataset had been loaded successfully.

5.2 Dataset Structure and Information

The shape, columns, and info() methods were used to understand the dataset's dimensions, column names, and data types.

The dataset contains 178 records and 14 columns, including 13 numerical predictor features and the target variable.

5.3 Missing Value Analysis

Missing values were checked using the isnull().sum() method.

The analysis showed that there were no missing values in the cleaned dataset. Therefore, no additional missing-value imputation was required.

5.4 Duplicate Record Analysis

Duplicate records were identified using the duplicated().sum() method.

No duplicate rows were found in the cleaned dataset, indicating that no duplicate-removal operation was necessary.

5.5 Descriptive Statistical Analysis

Descriptive statistics were calculated using the describe() method.

The analysis included the following statistical measures:

- Count
- Mean
- Standard deviation
- Minimum and maximum values
- First quartile (25%)
- Median (50%)
- Third quartile (75%)

These measures help explain the central tendency and variability of the numerical features.

5.6 Wine Class Distribution

A bar chart was created to visualize the number of samples belonging to each wine class.

The visualization showed that Class 1 had the highest sample count, followed by Class 0 and Class 2.

Understanding class distribution is useful when evaluating the dataset and planning subsequent classification tasks.

5.7 Feature Distribution Analysis

Histograms were created to examine the distributions of numerical features.

Histograms help identify the spread of values, possible skewness, and differences in the distributions of individual features.

The visualizations also provide an overview of how the numerical measurements vary across the dataset.

5.8 Correlation Analysis

A correlation heatmap was generated to examine the relationships between numerical variables.

Correlation coefficients range from -1 to +1:

- A positive correlation indicates that two variables tend to increase together.
- A negative correlation indicates that one variable tends to decrease as the other increases.
- A correlation close to zero indicates a weak linear relationship.

The heatmap provides an overview of the linear relationships between the numerical features. Correlation alone does not establish causation.

5.9 Outlier Detection Using IQR

Potential outliers were identified using the Interquartile Range method.

The calculations were performed using the following formulas:

IQR = Q3 − Q1

Lower Bound = Q1 − 1.5 × IQR

Upper Bound = Q3 + 1.5 × IQR

Values below the lower bound or above the upper bound were flagged as potential outliers.

The analysis was performed on numerical predictor features, excluding the target variable.

Potential outliers were identified for further examination rather than automatically removed. This approach helps preserve valid observations that may represent natural variation in the data.

5.10 Scatter Plot Analysis

A scatter plot was created to investigate the relationship between alcohol and proline.

The observations were displayed using different labels for the three wine classes.

This visualization helps examine whether the two features show different patterns across classes and whether they may be useful for distinguishing wine samples.

5.11 Class-Wise Alcohol Analysis

Alcohol content was summarized separately for each wine class.

The following measures were calculated:

- Number of samples
- Mean
- Median
- Standard deviation
- Minimum
- Maximum

These statistics allow a comparison of alcohol content and its variability across the three classes.

6. Statistical Hypothesis Testing

6.1 Objective

A one-way Analysis of Variance (ANOVA) test was performed to determine whether the mean alcohol content differed among the three wine classes.

6.2 Hypotheses

Null Hypothesis (H_0):

The population mean alcohol content is equal across all three wine classes.

Alternative Hypothesis (H_1):

At least one wine class has a different population mean alcohol content.

6.3 Significance Level

The significance level was set to:

$\alpha = 0.05$

6.4 ANOVA Results

The test produced the following results:

Statistical Measure	Result
F-statistic	135.0776
P-value	3.3195×10^{-36}
Significance Level	0.05

6.5 Interpretation

The calculated p-value is considerably smaller than the significance level of 0.05.

Therefore, the null hypothesis was rejected.

The result indicates a statistically significant difference in mean alcohol content among the three wine classes.

The F-statistic of 135.0776 provides evidence that the differences between the class means are large relative to the variation within the groups.

However, the ANOVA test does not identify which specific pairs of wine classes differ significantly. A suitable post-hoc test, such as Tukey's HSD, would be required to investigate pairwise differences.

The result demonstrates an association between wine class and alcohol content in this dataset. It does not, by itself, establish that alcohol content can accurately classify every wine sample.

7. Key Findings

The main findings of the analysis are summarized below:

1. The cleaned dataset contains 178 records and 14 columns.
2. The dataset includes 13 numerical predictor features and one target variable, wine_class.
3. The dataset contains three wine classes: Class 0, Class 1, and Class 2.
4. Class 1 contains the highest number of samples, with 71 records.
5. No missing values or duplicate records were found in the cleaned dataset.
6. Descriptive statistics provided an overview of the numerical features and their variability.
7. Histograms were used to examine feature distributions.
8. The correlation heatmap was used to explore linear relationships between numerical features.
9. The IQR method was used to flag potential outliers for further investigation.
10. The scatter plot helped explore the relationship between alcohol and proline across the wine classes.
11. The one-way ANOVA test produced an F-statistic of 135.0776 and a p-value of 3.3195×10^{-36}.
12. The ANOVA result indicates a statistically significant difference in mean alcohol content among the wine classes.

8. Challenges and Limitations

Several considerations are important when interpreting the results.

8.1 Outlier Interpretation

The IQR method identifies unusual values based on statistical boundaries. However, an observation flagged as an outlier is not necessarily incorrect. Potential outliers should be examined before deciding whether to retain or remove them.

8.2 Correlation Limitations

A correlation heatmap primarily describes linear relationships between numerical variables. It cannot establish causation and may not reveal every type of relationship.

8.3 ANOVA Limitations

One-way ANOVA evaluates whether the group means are equal under its assumptions. A significant result does not identify which specific classes differ. Post-hoc testing is needed for pairwise comparisons.

8.4 Exploratory Nature of the Analysis

EDA helps understand the dataset, but it does not establish the predictive performance of a machine learning model. A separate modeling task is required to evaluate classification performance.

9. Conclusion

In Week 2, Exploratory Data Analysis was performed on the Wine Recognition dataset to understand its structure, examine data quality, investigate numerical feature distributions, and explore relationships among variables.

The analysis included dataset inspection, missing-value and duplicate checks, descriptive statistics, wine class distribution visualization, histograms, a correlation heatmap, IQR-based potential outlier detection, and scatter plot analysis.

The cleaned dataset contained 178 records, 14 columns, and three wine classes. No missing values or duplicate records were identified.

A one-way ANOVA test was conducted to determine whether mean alcohol content differed among the three wine classes. The test produced an F-statistic of 135.0776 and a p-value of 3.3195×10^{-36}. Since the p-value was below the significance level of 0.05, the null hypothesis was rejected. This indicates a statistically significant difference in mean alcohol content among the wine classes.

Overall, the EDA process provided a better understanding of the dataset and identified alcohol content as a feature worth investigating further for wine-class differentiation. The findings establish a foundation for subsequent statistical analysis and machine learning modeling.

10. References

1. UCI Machine Learning Repository. *Wine Recognition Dataset*.
2. Scikit-learn Documentation. *Wine recognition dataset*.
3. Pandas Documentation. *Data analysis and manipulation*.
4. Matplotlib Documentation. *Visualization with Python*.
5. SciPy Documentation. *Statistical functions and hypothesis testing*.
6. Seaborn Documentation. *Statistical data visualization*.

End of Week 2 Report